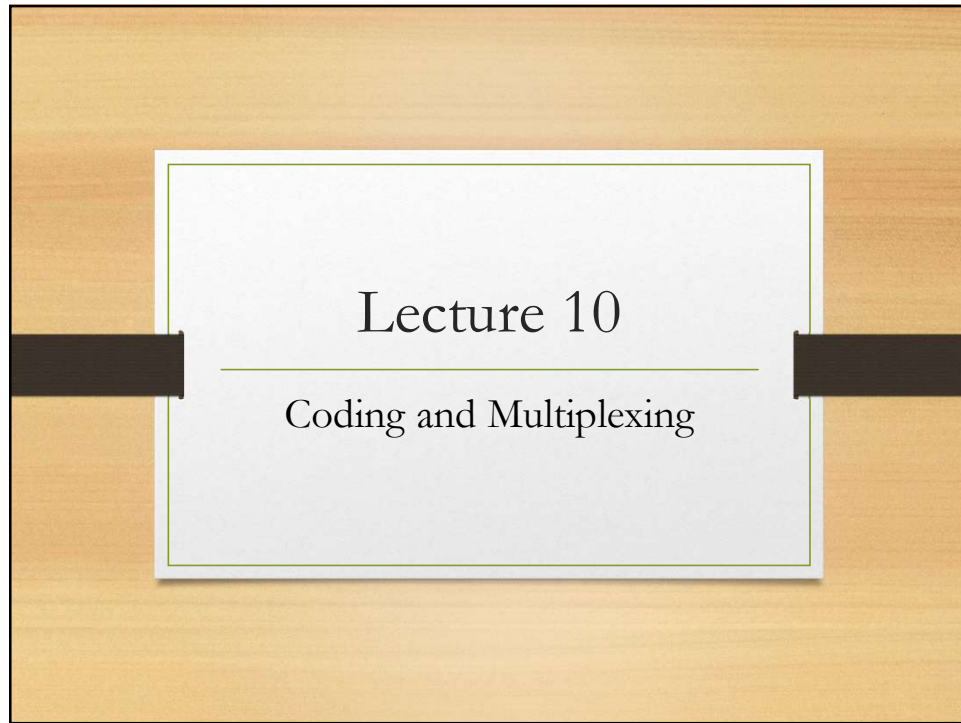
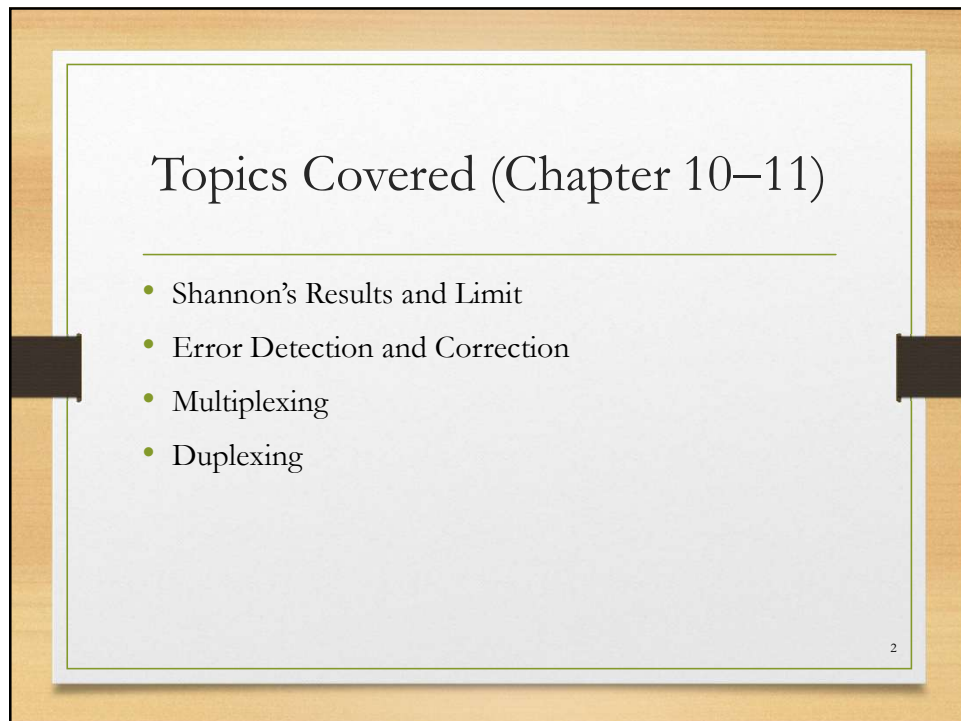




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Noisy Channel

- The entropy coding with aims to achieve zero redundancy cannot be used over a noisy channel.
 - Use of redundancy, in general, helps combat noise.
 - Example: single parity code—one extra bit is added to ensure each code word to have even or odd number of 1's.
- The addition of one extra bit reduces coding efficiency

$$\eta = \frac{H(m)}{H(m) + 1}$$

3

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Coding Redundancy

- Shannon has shown that it is possible to achieve error-free communication by adding sufficient redundancy.
- For example, if we have a binary symmetric channel (BSC) with finite error probability, then for error free communication over this channel, message from a source with entropy $H(m)$ must be encoded by binary codes with a word length at least $H(m)/C_s$, where $C_s (<1)$ is called the channel capacity.

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Per-Bit Transmission

- Consider a simple method to reduce error probability by repeating a given digit 3 times

0 → 000 1 → 111

At the receiver majority logic is used.

If two or more 0's are received, it declares 0 else 1.

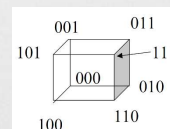
By using this code, we can have error-free communication if there is one error. This scheme will fail if there are two errors, for which we need five repetitions.

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Per-Bit Transmission

- By increasing the number of repetition, n , we can reduce the error probability even there are more number of bit errors. At the same time, the data rate will also be reduced by a factor of n (due to repetitions). However, no matter how large is n , the error probability will not be zero.
- This is because we are considering each bit.



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Group-Bits Transmission

- Suppose we need to transmit α digits per second, so over a period of T seconds, we have αT digits. To this group, we add $(\beta - \alpha)T$ check digits.
- That is equivalent to we occupy $2^{-(\beta - \alpha)T}$ fraction of the vertices of the βT dimension hyper-cube.
- In order to reduce the error probability, we can simply increase $T \rightarrow \infty$. But what is the maximum value of the ratio α/β for this dream to come true?

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Shannon's Results

- Shannon's results mentioned that

$$\frac{\alpha}{\beta} < C_s$$

where

$$C = B \log_2 \left(1 + \frac{S}{N} \right), \quad C_s = \frac{C}{K}$$

K is the symbol rate.

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Shannon's Results

- Firstly, Shannon's results indicate an upper limit on the rate of error-free communication that can be achieved.
- Secondly, it shows a way to reduce the error probability with a small reduction in the rate of transmission of information.

It allows us to seek a compromise between error-free communication with infinite delay and virtually error-free communication with finite delay.

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Shannon's Limit

- The capacity of a channel of infinite bandwidth

For white noise $N=N_0B$, where N_0 is the noise density, we can find out C when $B \rightarrow \infty$

$$\begin{aligned} \lim_{B \rightarrow \infty} C &= \lim_{B \rightarrow \infty} B \log_2 \left(1 + \frac{S}{N} \right) \\ &= \lim_{B \rightarrow \infty} \left\{ \frac{S}{N_0} \left[\frac{N_0 B}{S} \log_2 \left(1 + \frac{S}{N_0 B} \right) \right] \right\} \\ &= \frac{S}{N_0} \lim_{B \rightarrow \infty} \left[\frac{N_0 B}{S} \log_2 \left(1 + \frac{S}{N_0 B} \right) \right] \\ &= \frac{S}{N_0} \log_2 e = 1.44 S / N_0 \text{ bps} \end{aligned}$$

$S = E_b R, \quad C = 1.44 S / N_0 \Rightarrow C = 1.44 E_b R / N_0$
if $R = C$, then
 E_b / N_0
 $= 1 / 1.44$
 $\approx -1.6 \text{ dB}$
Shannon's Limit

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11-7: Error Detection and Correction

- When high-speed binary data is transmitted over a communication link, whether it is a cable or radio, errors will occur.
- These errors are changes in the bit pattern caused by interference, noise, or equipment malfunctions.
- Such errors will cause incorrect data to be received.
- The number of bit errors that occur for a given number of bits transmitted is referred to as the **bit error rate (BER)**.

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11-7: Error Detection and Correction

- The process of error detection and correction involves adding extra bits to the data characters to be transmitted. This process is generally referred to as **channel encoding**.
- The data to be transmitted is processed in a way that creates the extra bits and adds them to the original data. At the receiver, these extra bits help in identifying any errors that occur in transmission caused by noise or other channel effects.

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11-7: Error Detection and Correction

- A key point about channel encoding is that it takes more time to transmit the data because of the extra bits. These extra bits are called **overhead** in that they extend the time of transmission.
- Channel encoding methods fall into two separate categories, **error detection codes** and **error correction codes**.

13

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Error Detection

- Many different methods have been used to ensure reliable error detection:
 - **Redundancy** is a method that ensures error-free transmission by sending each character or message multiple times until it is properly received.
 - **Encoding schemes** like the RZ-AMI are used whereby successive binary 1 bits in the bit stream are transmitted with alternating polarity.

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Error Detection

- One of the most widely used systems is known as **parity**, in which each character transmitted contains one additional bit, known as a **parity bit**.
- The **cyclical redundancy check (CRC)** is a mathematical technique used in synchronous data transmission that effectively catches 99.9 percent or more of transmission errors.

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Error Correction

- A number of efficient error-correction schemes have been devised to complement error detection methods.
- The process of detecting and correcting errors at the receiver so that retransmission is not necessary is called **forward error correction (FEC)**.
- There are two basic types of FEC: **block codes** and **convolutional codes**.

16

16

Error-Correction Codes

- **Block Codes**—A block of k data bits is encoded by a code word of n bits ($n > k$). For each k sequence of bits there is a unique n bit word.
- **Convolutional Codes**—Data bits are continuously generated. The output bits not only depend on the present bit but also on few earlier bits.

17

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Block Codes

- The **block check character (BCC)** is also known as a **horizontal** or **longitudinal redundancy check (LRC)**.
- It is the process of logically adding, by exclusive-ORing, all the characters in a specific block of transmitted data.
- The final bit value for each horizontal row becomes one bit in a character known as the **block-check character (BCC)**, or the **block-check sequence (BCS)**.

18

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Block Codes

- The most popular FEC codes are the **Hamming** and **Reed Solomon** codes.
- These codes add extra parity bits to a transmitted word, process them using unique algorithms, and detect and correct bit errors.
- **Interleaving** is a method used in wireless systems to reduce the effects of burst errors.

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Convolutional Codes

- Convolutional encoding creates additional bits from the data as do Hamming and Reed Solomon codes, but the encoded output is a function of not only the current data bits but also previously occurring data bits.
- Convolutional codes pass the data to be transmitted through a special shift register.
- As the serial data is shifted through the shift register flip-flops, some of the flip-flop outputs are XORed together to form two outputs.

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Convolutional Codes

- These two outputs are the convolutional code, and this is what is transmitted.
- The original data itself is not transmitted.
- Instead, two separate streams of continuously encoded data are sent.
- Since each output code is different, the original data can more likely be recovered at the receiver by an inverse process.

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Convolutional Codes

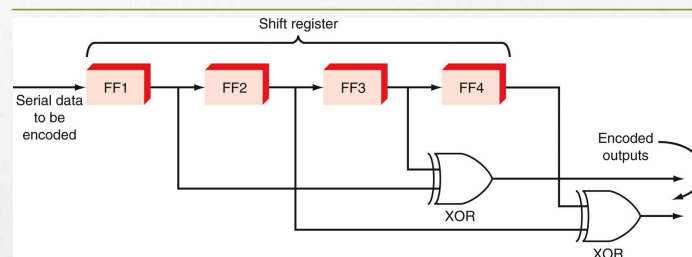


Figure 11-56: Convolutional encoding uses a shift register with exclusive-OR gates to create the output.

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10-1: Multiplexing Principles

- Transmitting two or more signals simultaneously can be accomplished by running multiple cables or setting up one transmitter-receiver pair for each channel, but this is an expensive approach.
- A single cable or radio link can handle multiple signals simultaneously using a technique known as **multiplexing**.
- Multiplexing permits hundreds or even thousands of signals to be combined and transmitted over a single medium.

23

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10-1: Multiplexing Principles

- **Multiplexing** is the process of simultaneously transmitting two or more individual signals over a single communication channel.
- It increases the number of communication channels so that more information can be transmitted.
- An application may require multiple signals.
- Cost savings can be gained by using a single channel to send multiple information signals.

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10-1: Multiplexing Principles

- Four communication applications that would be prohibitively expensive or impossible without multiplexing are:
 1. Telephone systems
 2. Telemetry
 3. Satellites
 4. Broadcasting (radio and TV)

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10-1: Multiplexing Principles

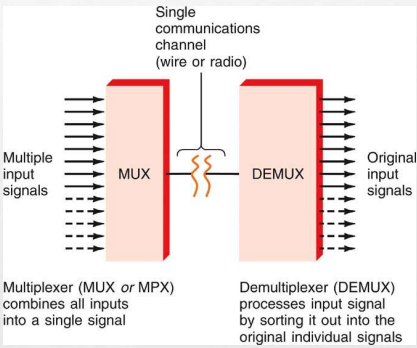


Fig. 10-1: Concept of multiplexing.

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10-1: Multiplexing Principles

- The two most common types of multiplexing are
 1. Frequency-division multiplexing (FDM)
 - Generally used for analog information.
 - Individual signals to be transmitted are assigned a different frequency within a common bandwidth.
 2. Time-division multiplexing (TDM)
 - Generally used for digital information.
 - Multiple signals are transmitted in different time slots on a single channel.

27

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10-1: Multiplexing Principles

- Another form of multiple access is known as **code-division multiple access (CDMA)**.
 - Widely used in cell phone systems to allow many subscribers to use a common bandwidth simultaneously.
 - Uses special codes assigned to each user that can be identified.

28

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10-2: Frequency-Division Multiplexing

- In frequency-division multiplexing (FDM), multiple signals share the bandwidth of a common communication channel.
- All channels have specific bandwidths.
- A wide bandwidth can be shared for the purpose of transmitting many signals at the same time.

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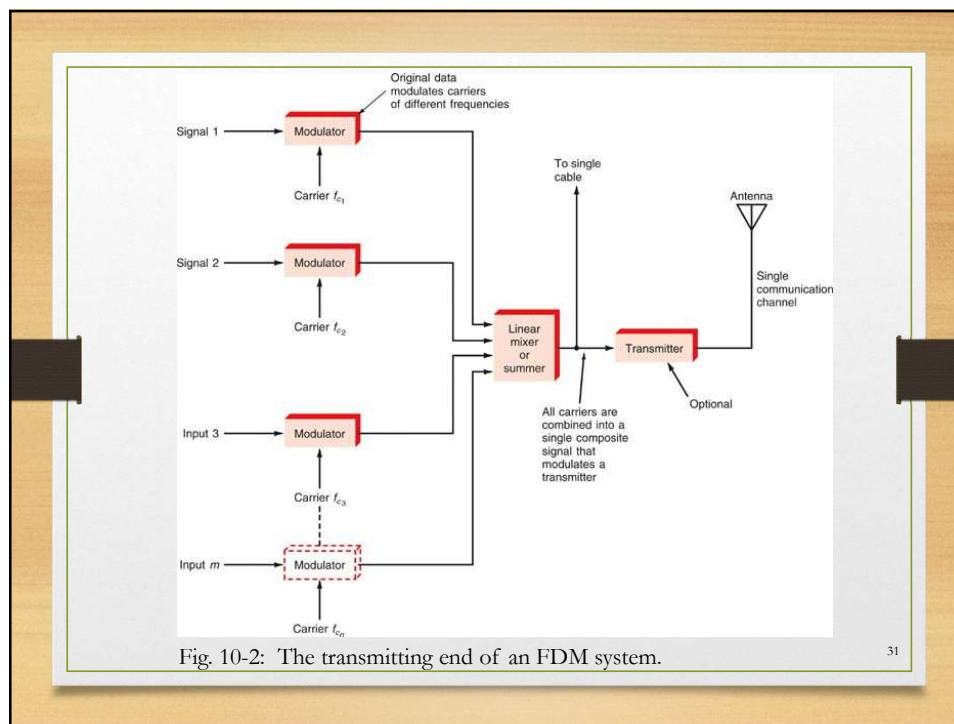
10-2: Frequency-Division Multiplexing

Transmitter-Multiplexers

- In an FDM system, each signal to be transmitted feeds a modulator circuit.
- The carrier for each modulator (f_c) is on a different frequency.
- The carriers are equally spaced from one another.
- These carriers are referred to as **subcarriers**.
- Each input signal is given a portion of the bandwidth.
- The FDM process divides up the bandwidth of the single channel into smaller, equally spaced channels, each capable of carrying information in sidebands.

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10-2: Frequency-Division Multiplexing

Transmitter-Multiplexers

- The modulator outputs containing the sideband information are added algebraically in a linear mixer.
- The resulting output signal is a composite of all the modulated subcarriers.
- This signal can be used to modulate a radio transmitter, or can itself be transmitted over a single channel.
- The composite signal can also become one input to another multiplexed system.

32

32

10-2: Frequency-Division Multiplexing

Receiver-Demultiplexer

- In an FDM system, a receiver picks up the signal and demodulates it, recovering the composite signal.
- The composite signal is sent to a group of bandpass filters, each centered on one of the carrier frequencies.
- Each filter passes only its channel and rejects all others.
- A channel demodulator then recovers each original input signal.

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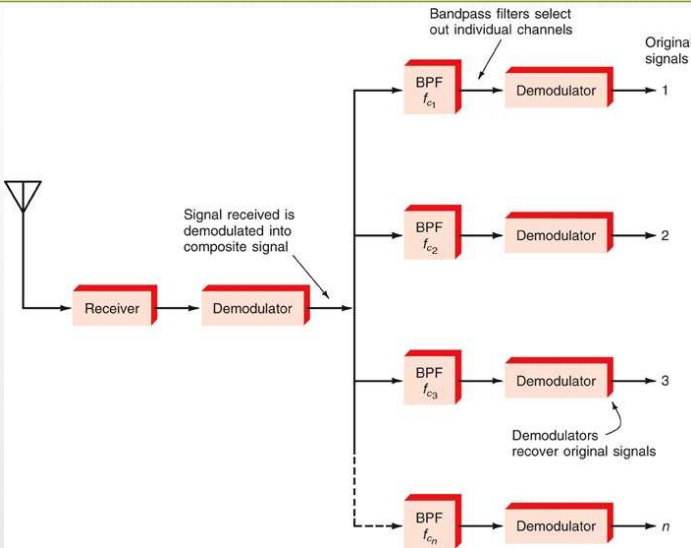


Figure 10-4: The receiving end of an FDM system.

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10-2: Frequency-Division Multiplexing

FDM Applications: Telemetry

- Sensors in telemetry systems generate electrical signals that change in some way in response to changes in physical characteristics.
- An example of a sensor is a **thermistor**, a device used to measure temperature.
- A thermistor's resistance varies inversely with temperature.

35

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10-2: Frequency-Division Multiplexing

FDM Applications: Telemetry

- The thermistor is usually connected into a resistive network, such as a voltage divider or bridge.
- The thermistor is also connected to a DC voltage source.
- The result is a DC output voltage which varies in accordance with temperature.
- This voltage is transmitted to a remote receiver for measurement, readout, and recording.
- The thermistor becomes one channel of an FDM system.

36

36

10-2: Frequency-Division Multiplexing

FDM Applications: Telemetry

- The varying direct or alternating current changes the frequency of an oscillator operating at the carrier frequency.
 - Such a circuit is called a **voltage-controlled oscillator (VCO)** or **subcarrier oscillator (SCO)**.
- Most VCOs are astable multivibrators whose frequency is controlled by the input from the signal conditioning circuits.
- A system that uses FM of the VCO subcarriers as well as FM of the final carrier is called **FM/FM**.

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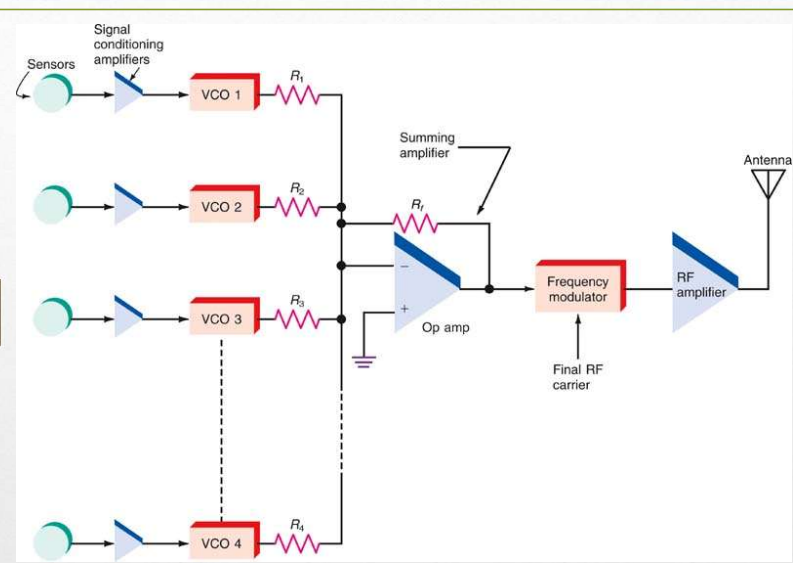
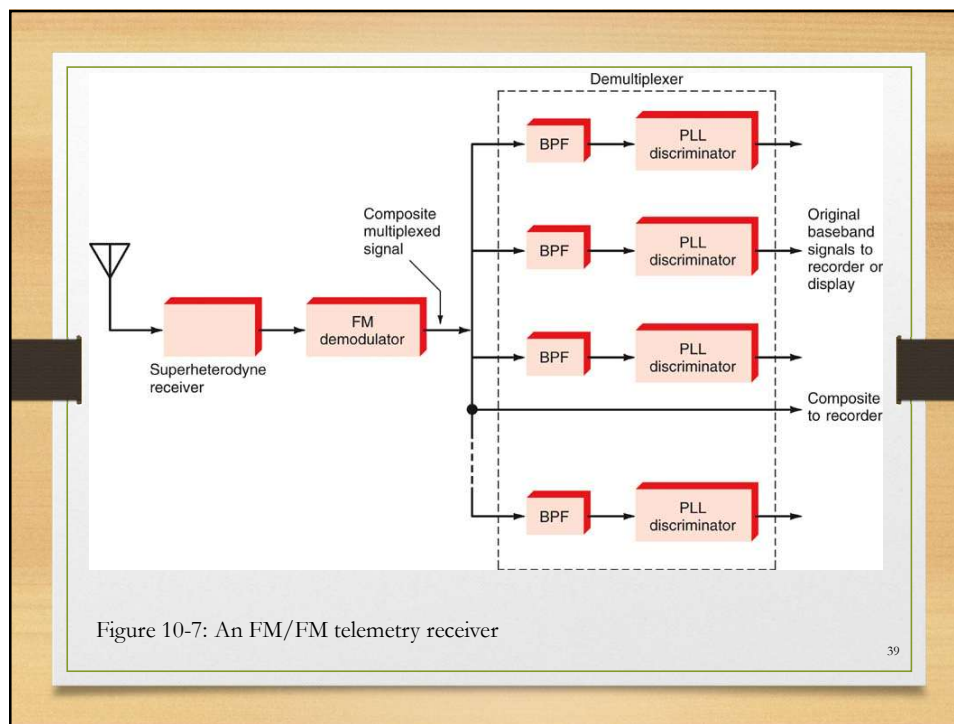


Figure 10-5: An FDM telemetry transmitting system.

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10-2: Frequency-Division Multiplexing

FDM Applications: Telemetry

- On the receiving end of a telemetry system, an FM demodulator reproduces the original composite multiplexed signal, which is then fed to a demultiplexer.
- The demultiplexer divides the signals and reproduces the original inputs.
- The output of the first FM demodulator is fed simultaneously to multiple bandpass filters, each of which is tuned to the center frequency of one of the specified subchannels.

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10-2: Frequency-Division Multiplexing

FDM Applications: Telemetry

- Each filter passes only its subcarrier and related sidebands and rejects all the others.
- The demultiplexing process essentially uses filters to sort the composite multiplex signal back into its original components.
- The output of each filter is the subcarrier oscillator frequency with its modulation.

41

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10-2: Frequency-Division Multiplexing

FDM Applications: Telemetry

- These signals are then applied to FM demodulators. Also known as **discriminators**, these circuits take the FM signal and recreate the original dc or ac signal produced by the transducer.
- The original signals are measured or processed to provide the desired information from the remote transmitting source.
- In most systems, the multiplexed signal is sent to a data recorder where it is stored for possible future use.

42

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10-2: Frequency-Division Multiplexing

FDM Applications: Telephone Systems

- For decades, telephone companies used FDM to send multiple telephone conversations over a minimum number of cables.
- The original voice signal, in the 300- to 3000-Hz range is used to modulate a subcarrier.
- Lower sideband (LSB) SSB AM was used.
- Each subcarrier is on a different frequency. Those subcarriers are added together to form one channel.
- The FDM system has been replaced by an all-digital time multiplexing (TDM) system.

43

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10-2: Frequency-Division Multiplexing

FDM Applications: Cable TV

- In a cable TV system, TV signals, each in its own 6-MHz channel, are multiplexed on a common coaxial or fiber-optic cable and sent to homes.
- Each 6-MHz channel carries the video and voice of the TV signal.
- Coaxial and fiber-optic cables have an enormous bandwidth and can carry more than one hundred TV channels.
- Many cable TV companies also use their cable system for Internet access.

44

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10-2: Frequency-Division Multiplexing

FDM Applications: FM Stereo Broadcasting

- In recording original stereo, two microphones are used to generate two separate audio signals.
- Two microphones pick up sound from a common source, such as voice, but from different directions.
- The separation of the two microphones provides sufficient differences in the two audio signals to provide a realistic reproduction of the original sound.
- FDM techniques are used to transmit these independent signals by a single transmitter.

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10-3: Time-Division Multiplexing

- In FDM, multiple signals are transmitted over a single channel, each signal being allocated a portion of the spectrum within that bandwidth.
- In **time-division multiplexing (TDM)**, each signal occupies the entire bandwidth of the channel.
- Each signal is transmitted for only a brief period of time.
- TDM can be used with both digital and analog signals.

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10-3: Time-Division Multiplexing

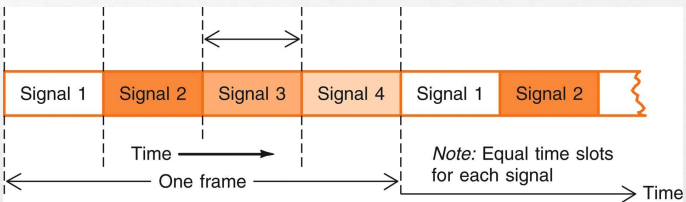


Figure 10-12: The basic TDM concept.

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10-3: Time-Division Multiplexing

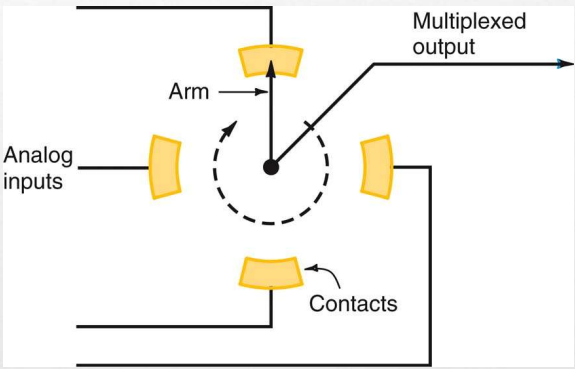


Figure 10-14: Simple rotary-switch multiplexer.

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10-5: Duplexing

- **Duplexing** is the method by which two-way communications are handled.
- Half duplexing means that the two stations communicating take turns transmitting and receiving.
- Full duplexing means that the two stations can send and receive simultaneously.
- There are two ways to provide duplexing:
 - Frequency-division duplexing (FDD)
 - Time-division duplexing (TDD).

49

49

10-5: Duplexing

- FDD is the simplest way to provide full duplex.
- FDD uses two separate channels, one for sending and another for receiving.
- The big disadvantage of this method is the extra spectrum space required.
- However, most cell phone systems use this method because it is the easiest to implement and the most reliable.

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10-5: Duplexing

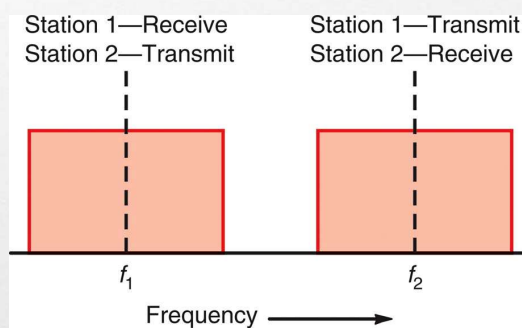


Figure 10-27: Frequency-division duplexing (FDD).

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10-5: Duplexing

- Time-division duplexing (TDD) means that signals are transmitted simultaneously on a single channel by interleaving them in different time slots.
- Each time slot may contain one data word, such as 1 byte from an A/D converter or a D/A converter.
- The primary benefit of TDD is that only one channel is needed, saving spectrum space and cost.

52

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10-5: Duplexing

- The TDD method is harder to implement.
- The key to making it work is precise timing and synchronization between transmitter and receiver.
- Special synchronizing pulses or frame sequences are needed to constantly ensure that timing will not result in collisions between transmit and receive.
- Several of the newer third-generation cell phone systems may use TDD.

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10-5: Duplexing

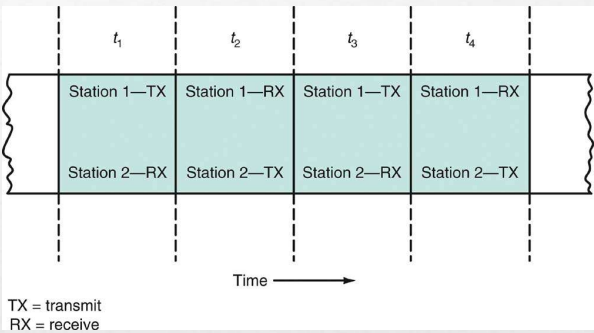


Figure 10-28: Time-division duplexing (TDD).

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Line Codes

- Line codes provide a means of electrically representing a bit stream in digital baseband communication systems.
- Type of Line Codes:
 - NRZ codes: Non-Return-to-Zero-Level, unipolar and bipolar
 - RZ codes: Return-to-Zero, Alternate Mark Inversion

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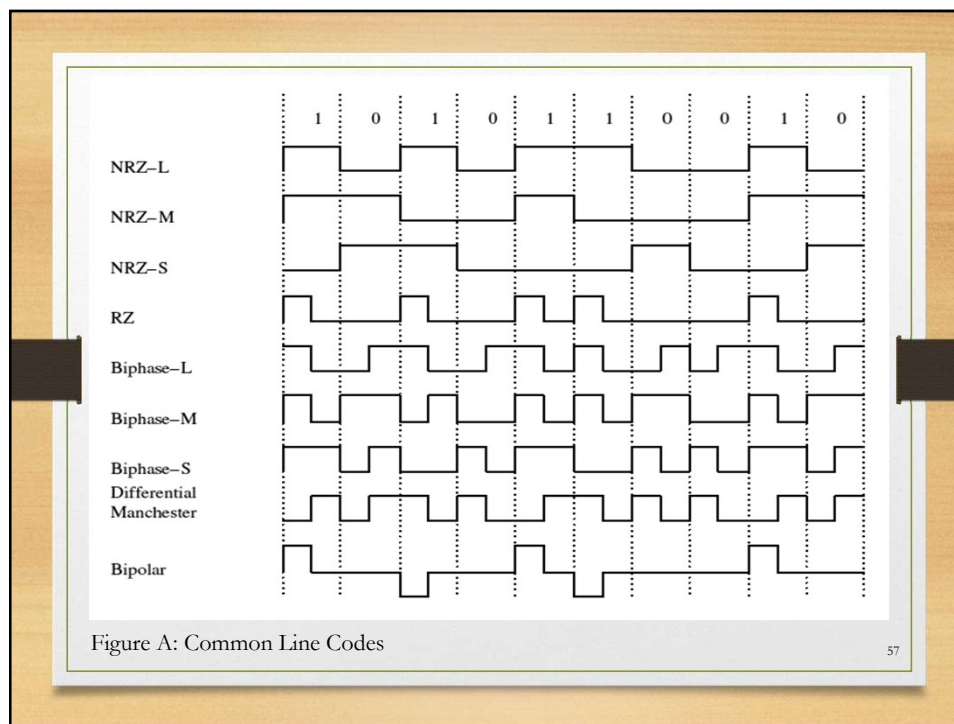
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Line Codes

- Type of Line Codes:
 - Biphasic Codes: Biphasic-L Code (Manchester Code), Biphasic-M Code, Biphasic-S code.
 - RLL Codes: Run-Length-Limited (RLL): Bipolar with n Zeros substituted (BnZS), Coded Mark Inversion, high density bipolar with 3 zero maximum (HDB3) code

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Pulse Shaping

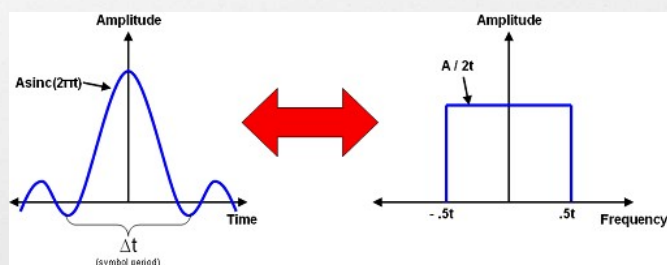
- Problem with Rectangular Pulse
 - High frequency components
 - Inter-symbol interference
- Required pulse shape
 - Each received pulse $p(t)$ must be shaped in a such way that although it spreads significantly into subsequent bit intervals, it always has zero value at the crucial sampling instants of those bit intervals.

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Pulse Shaping

- The sinc function is an ideal candidate for the required pulse shape.



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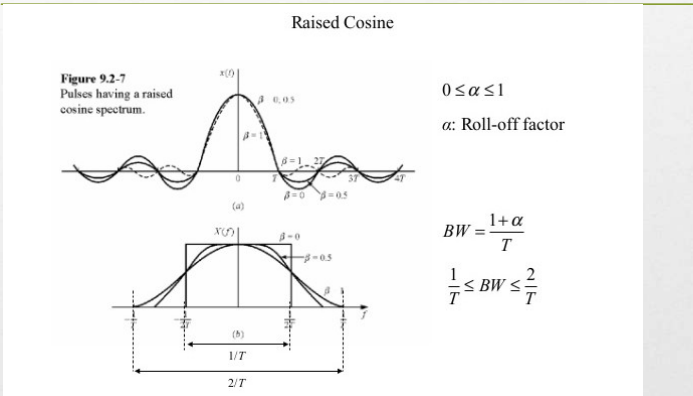
Sinc Function

- Two practical problems:
 - Ideal low cut-off characteristic in frequency domain is impossible to realise.
 - Required perfect timing, otherwise the accumulated residual energy from adjacent sinc pulses will contribute to decoding errors.
- Solution
 - Use raised cosine filtering to realise gradual roll-off in the frequency domain.

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Raised Cosine Filtered Pulse



61